7.4 PROBLEMS

1. Develop a mixed integer programming model for the following reservoir design problem and solve on LINGO.

   A farmer plans to develop water for irrigation. He is considering two possible sources of water: a gravity diversion from a possible reservoir with two alternative capacities and/or a pump from a lower river diversion (refer to Figure 7.3). Between the reservoir and pump site the river base flow increases by 2 acft/day due to groundwater drainage into the river. Ignore losses from the reservoir. The river flow into the reservoir and the farmer's demand during each of two six-month seasons of the year are given in Table 7.5. Revenue is estimated at $300 per year per acre irrigated.

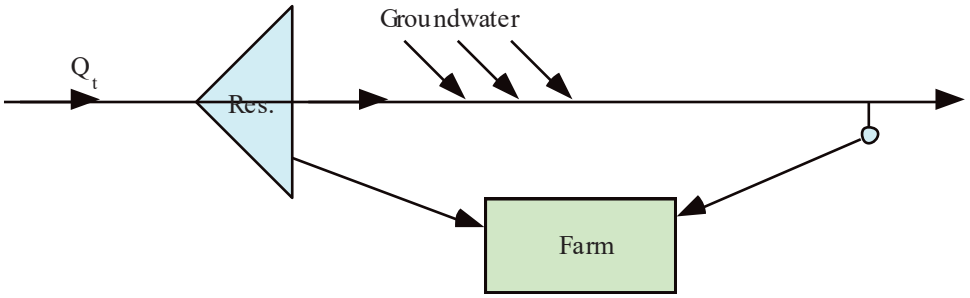


Figure 7.3: MIP and Irrigation Supply from a Reservoir and/or River Diversion

Table 7.5: Seasonal Flow and Demand

Season, t	River Inflow, Q_t (acft)	Irrigation Demand (acft/acre)
1	600	1.0
2	200	3.0

   Assume that there are only two possible sizes of reservoir: a high dam that has capacity of 700 acft or a low dam with capacity of 300 acft. The capital costs are $10,000/year and $6,000/year for the high and low dams, respectively (no operating cost). The pump capacity is fixed at 2.2 acft/day with a capital cost (if it is built) of $8,000/year and operating cost of $20/acft.

   Develop a mixed integer programming model for this problem and solve on LINGO.